1. Develop following system. Apply necessary constraints and create tables. Assume suitable data and fields.
   Users (user_id, name, age, gender, username)
   Websites(website_id, name, type, size)
   Access(user_id, website_id, date of access, time of access)
   Store information about users of the internet as name, age, gender, username and their access to different websites with information as url, type of website as e.g. educational, entertainment etc and size of website as small or medium or large. System should store information about date of access of different websites by different users.
      Display names of users starting with letter S.
      Display number of websites in each type.
      Update age of the user "Geeta" from 20 to 21
      Display information of all male users
      Write suitable trigger.

2. Develop following system. Apply necessary constraints and create tables. Assume suitable data and fields.
   Users (user_id, name, age, gender, username)
   Websites(website_id, name, type, size)
   Access(user_id, website_id, date of access, time of access)
   Store information about users of the internet as name, age, gender, username and there access to different websites with information as url, type of website as e.g. educational, entertainment etc. and size of website as small or medium or large. System should store information about date of access of different websites by different users.
      Display names of users ending with letter R.
      Create view to display all websites accessed so far.
      Update age of the user "Geeta" from 20 to 21
      Display information of all female users
      Write suitable trigger

3. Develop following system. Apply necessary constraints and create tables. Assume suitable data and fields.
   Bill (bill_no, date, totalamount)
   Billdetails (bill_no, itemname, price, quantity)
   Store information about bill in typical restaurant showing details as bill no, bill date, items, quantity, rate, bill amount etc.
      Display total bill amount on a particular date.
      Modify bill date in one of the bill.
      Display bill details of particular bill_no
      Create trigger for taking backup of bill information before deletion of any bill.
      Write suitable stored procedure.

4. Develop following system. Apply necessary constraints and create tables. Assume suitable data and fields.
   Bill (<u>bill_no</u>, date, totalamount)
   Billdetails (<u>bill_no, itemname</u>, price, quantity)
   Store information about bill in typical restaurant showing details as bill no, bill date, items, quantity, rate, bill amount etc.
      Display total bill amount on a particular date.
      Modify bill date in one of the bill.
      Display bill details of particular bill_no
      Create trigger for taking backup of bill information before deletion of any bill.
      Write suitable stored procedure.

5. Create table/tables and apply necessary constraints. Assume suitable data.
   Patient(<u>Pid</u>, name, gender,DOB, address, telephoneno, bloodgroup)
   Visit (Pid, dateofvisit, diagnosis, medicine, feespaid)
   Store information about patients such as patient details as name, gender, DOB, address, telephone number, blood group, diagnosis during each visit, medicine prescription given in each visit, fees paid in each visit etc.
      Alter data type of telephone no. from int to bigint
      Display number of patients of each blood group
      Display patient names and their total visits made.
      Create view to display name and address of each patient.
      Create trigger to take backup of patients

6. Create table/tables and apply necessary constraints. Assume suitable data.
   Patient(<u>Pid</u>, name, gender,DOB, address, telephoneno, bloodgroup)
   Visit (Pid, dateofvisit, diagnosis, medicine, feespaid)
   Store information about patients such as patient details as name, gender, DOB, address, telephone number, blood group, diagnosis during each visit, medicine prescription given in each visit, fees paid in each visit etc.
      Alter data type of telephone no. from int to bigint
      Display number of patients of each blood group
      Display patient names and their total visits made.
      Create view to display name and address of each patient.
      Create trigger to take backup of patients

7. For University database execute following queries:
      Department (<u>dept_name</u>, building, budget)
      Instructor (<u>inst_id</u>, name, salary, dept_name)
      Course (<u>course_id</u>, title, credits, dept_name)
      Teaches (course_id, inst_id)
         Find the names of all departments whose name starts with " p ".
         List the entire instructor relation in descending order.
         Find the names of all instructors whose salary is between 20000 and 30000
         Delete information of particular instructor
         Create view of instructor name & course for instructors in IT department.

8. For University database execute following queries:

   Department (dept_name, building, budget)

   Instructor (inst_id, name, salary, dept_name)

   Course (course_id, title, credits, dept_name)

   Teaches (course_id, inst_id)

   Find the names of all departments whose name starts with " p ".

   List the entire instructor relation in descending order.

   Find the names of all instructors whose salary is between 20000 and 30000

   Delete information of particular instructor

   Create view of instructor name & course for instructors in IT department.

9. Create table/tables and apply necessary constraints. Assume suitable data.

   Patient(Pid, name, gender, DOB, address, telephoneno, bloodgroup)

   Visit (Pid, dateofvisit, diagnosis, medicine, feespaid)

   Store information about patients such as patient details as name, gender, DOB, address, telephone number, blood group, diagnosis during each visit, medicine prescription given in each visit, fees paid in each visit etc.

   Display details of all patients

   Add column email_id in the Patient table

   Display patient names and their total visits made.

   Display patient with age > 20 and <40.

   Create view to display name and blood group of male patients

10. Create table/tables and apply necessary constraints. Assume suitable data.

    Patient(Pid, name, gender, DOB, address, telephoneno, bloodgroup)

    Visit (Pid, dateofvisit, diagnosis, medicine, feespaid)

    Store information about patients such as patient details as name, gender, DOB, address, telephone number, blood group, diagnosis during each visit, medicine prescription given in each visit, fees paid in each visit etc.

    Display details of all patients

    Add column email_id in the Patient table

    Display patient names and their total visits made.

    Display patient with age > 20 and <40.

    Create view to display name and blood group of male patients

11. Create table/tables emp and dept with the following columns and apply necessary constraints. Assume suitable data.

    Emp- Emp_id, Ename, desg, qual, Salary, mob_no, Dept_id.

    Dept- Dept_id, Dname, Loc

    Find the names of all employees whose name starts with 'Sa'.

    List all the employees name and salary whose age is less than 40 years.

    Select the employees whose salary is between Rs. 20000 and Rs. 30000.

    Display average salary of all employee

    Write a function to calculate salary

12. Create table/tables emp and dept with the following columns and apply necessary constraints. Assume suitable data.
    Emp- Emp_id, Ename, desg, qual, Salary, mob_no, Dept_id.
    Dept- Dept_id, Dname, Loc
    - Find the names of all employees whose name starts with ‘Sa’.
    - List all the employees name and salary whose age is less than 40 years.
    - Select the employees whose salary is between Rs. 20000 and Rs. 30000.
    - Display average salary of all employee
    - Write a function to calculate salary

13. Draw an ER diagram for Book shop management system convert it into tables and Create atleast 2 tables in SQL with required constraints

14. Draw an ER diagram for Hostel management system convert it into tables and Create atleast 2 tables in SQL with required constraints

15. Draw an ER diagram for Hotel management system convert it into tables and Create atleast 2 tables in SQL with required constraints

16. Draw an ER diagram for Library management system convert it into tables and Create atleast 2 tables in SQL with required constraints